Transit Desert Analysis: City of Cape Town

An interpretation of the maps produced for each public transport mode and for combined networks. All maps use a 500 m walking distance buffer to represent coverage; bivariate classes are based on terciles of each variable.

1 Overview of Coverage and Key Patterns

Several important caveats apply to this interpretation:

- **Tercile-based classification:** Bivariate classes are computed independently per mode, so colour class boundaries are not directly comparable across maps.
- **Proximity-based measure:** The coverage represents potential walking access, not realised accessibility. Factors such as service frequency, transfers, fare costs, reliability, and safety are not captured.
- **Data uncertainty:** Golden Arrow bus stop locations were derived from on-board survey tracking and show dense point clusters in places. This may overestimate actual stop coverage.
- **Spatial gaps:** THe survey data does not cover the full metropole and a small number of suburbs present in the survey could not be fully mapped; however, broad spatial patterns remain clear.

1.1 Transit Need

The distribution of composite transit need (Figure 1) shows the highest need clustered in and around the Cape Flats (from Delft and Langa, southward to Pelikan Park, and across to Khayelitsha). These areas correspond to communities with higher poverty and lower private vehicle ownership, and larger shares of children and/or pension-receiving households.

2 Modal Comparison of Coverage

2.1 Trains

Rail coverage (Figure 2) is concentrated along the existing rail corridors. While parts of Khayelitsha align with the rail network and show high need with relatively high coverage, train stops are sparse across much of the northern Cape Flats where need is also very high. As a result, many high-need suburbs in those areas have limited rail access.

2.2 MyCiTi Buses

The MyCiTi network (Figure 2) provides dense coverage around the central city and extends northwards towards Parklands. It also covers Atlantis. However, it has limited penetration into much of the Cape Flats. Several high-need suburbs in these areas fall into the high-need, low-coverage category for MyCiTi.

2.3 Golden Arrow Buses

Golden Arrow (Figure 2) provides broader coverage across many high-need areas of the Cape Flats compared to MyCiTi and rail.

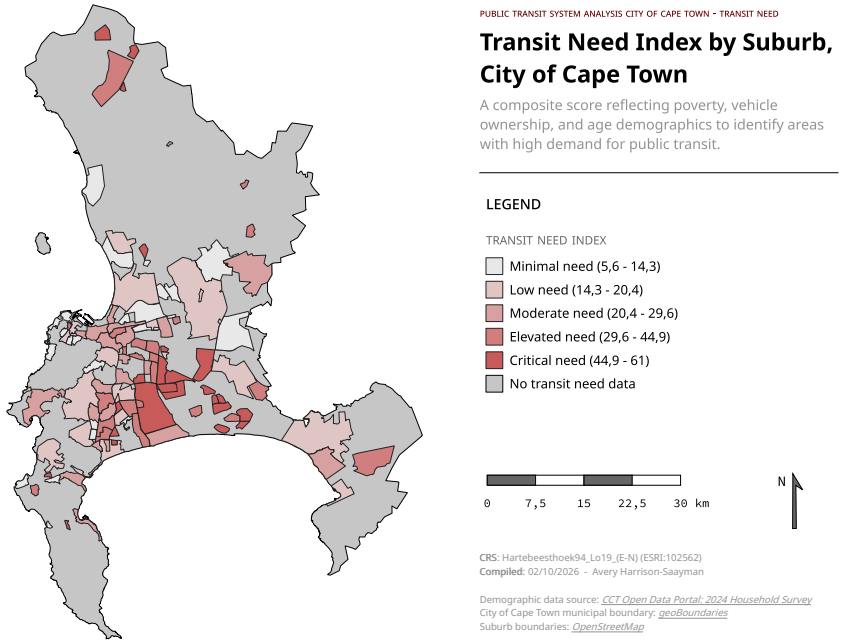


Figure 1: Transit Need

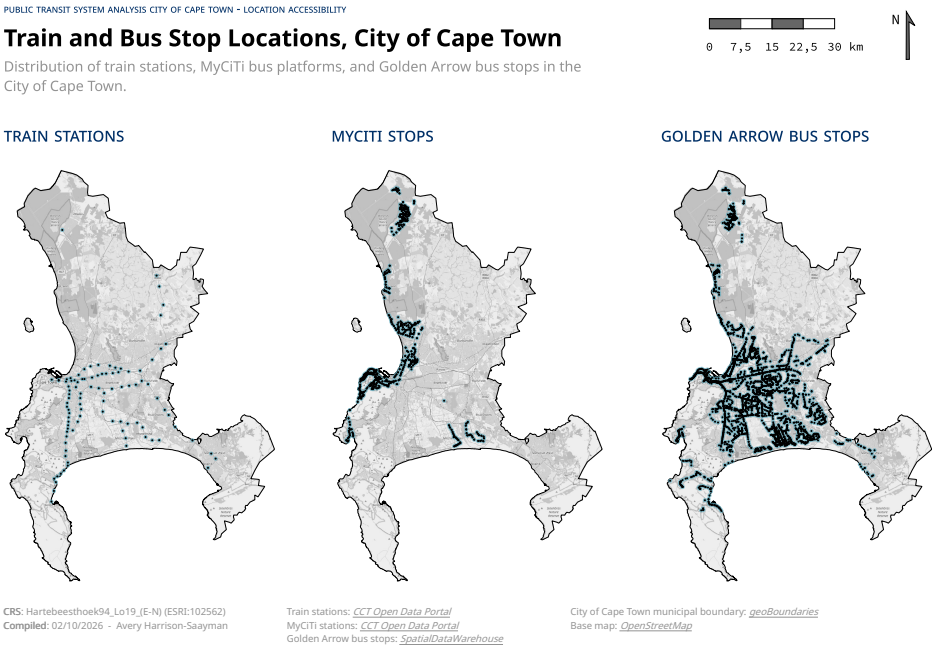


Figure 2: Train and bus stop coverage

2.4 Minibus Taxis

Taxis (Figure 3) form an extensive network across the metro. When taxi coverage is considered alongside other modes, many high-need areas show improved access. The route-based buffering captures the extent of the taxi network and suggests that taxis play a critical role in filling gaps left by formal scheduled services.

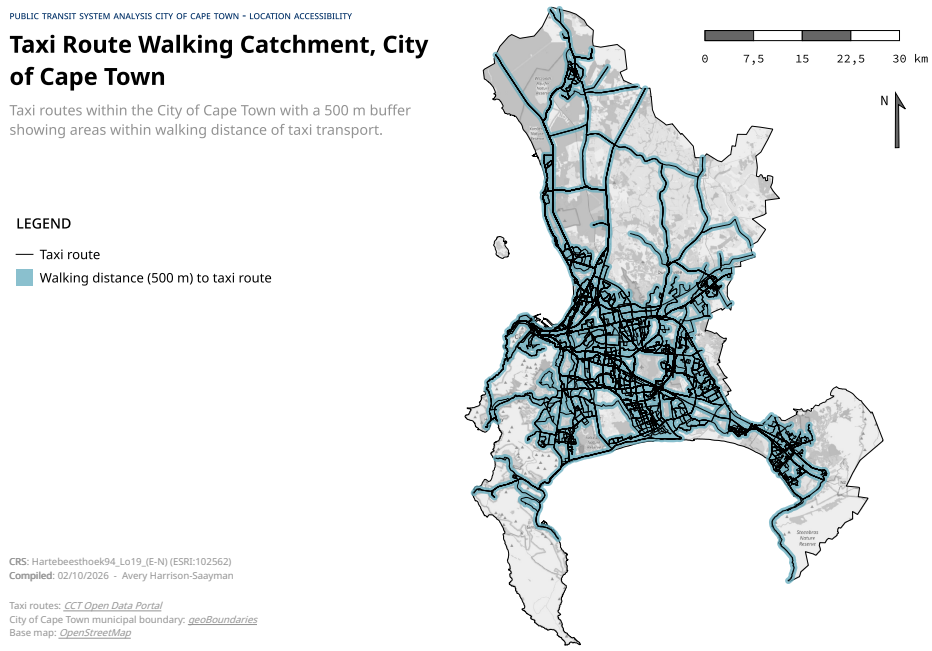


Figure 3: Train and bus stop coverage

2.5 Considering Coverage Against Need

Considering the bivariate choropleth map for a single mode, for example the rail network (Figure 4), reinforces the information that can be found in the other maps. For example, the areas of the Cape Flats identified as having a high need and where train stops are sparse are classified in the "high-need, low-coverage" and "high-need, medium-coverage" categories.

Combining modes and including all formal and informal modes (Figure 5) illustrates how network integration affects the classification of transit deserts. The extent of high-need, low-coverage areas is reduced relative to considering only single modes.

3 Identified Transit Deserts

A consistent set of suburbs emerge as transit deserts (high need, low coverage) across multiple mode configurations:

- **Pelikan Park** - High need with limited coverage, particularly for rail and MyCiTi.
- **Philippi** - Appears as a large area of high need with low coverage in several maps. It is worth noting that much of Philippi consists of smallholdings and farms, so population density is lower than in more built-up areas.

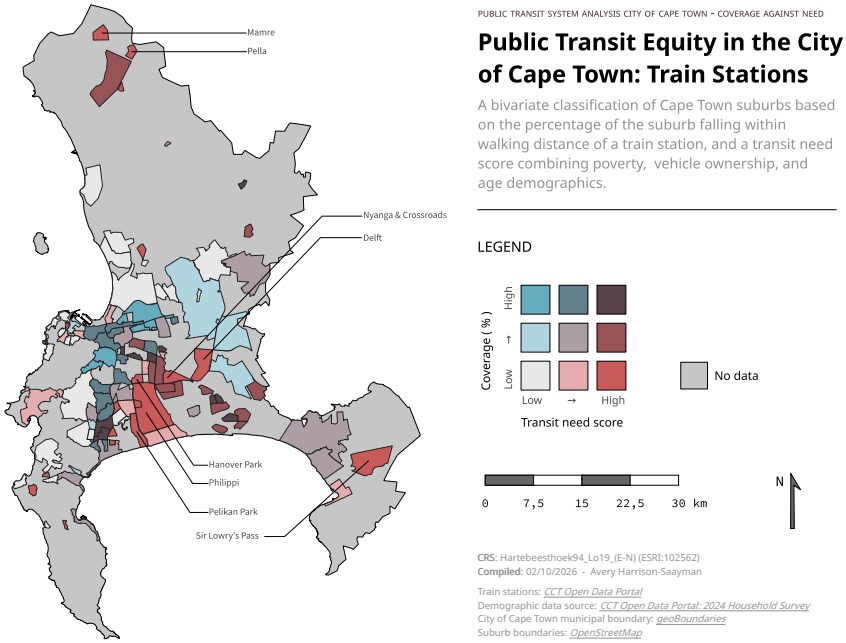


Figure 4: Transit Need vs Train Coverage

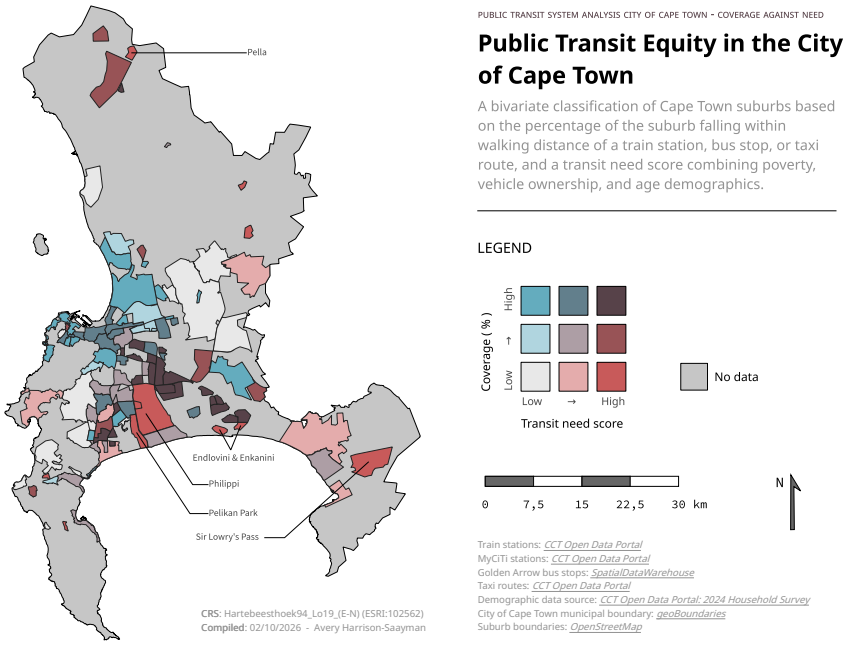


Figure 5: Transit Need vs All Public Transit Coverage

- **Enkanini and Endlovini (Southern Khayelitsha)** - High need and under-served relative to northern parts of Khayelitsha. The MyCiTi and rail services do not extend as comprehensively into these southern areas.

- **Sir Lowry's Pass** - Peripheral location with persistent low coverage relative to its need level.
- **Pella** - Consistently appears in the high-need, low-coverage class.

Within Khayelitsha, there is a clear intra-areal contrast: northern portions exhibit high need but high coverage (benefiting from a MyCiTi route and the train line), while southern portions (Enkanini, Endlovini) remain under-served.